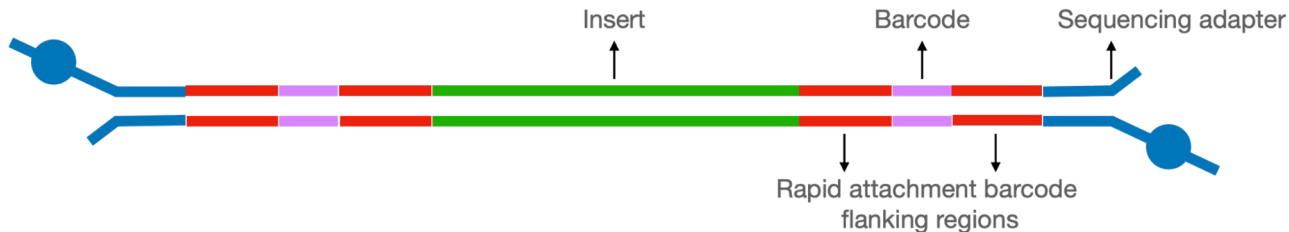


Library QC for ONT metagenomics

Library Quality Control (QC)

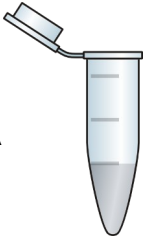
- The purpose of QC is to ensure that there is sufficient and good quality library for sequencing.
- Quantity (concentration)
 - Amount of library is measured using Qubit (or other fluorescence based method)
- Quality (length)
 - Measured using capillary electrophoresis (Tapestation, Bioanalyzer, Fragment Analyzer)
 - If capillary electrophoresis is unavailable, regular gel electrophoresis may be used with a diagnostic PCR



Overview of Library QC

QC 1

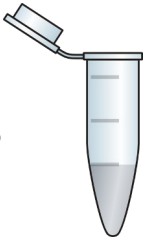
Sample A



Individual library QC passed?

YES

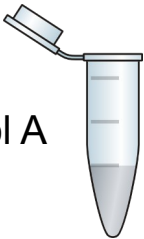
Sample B



Individual library QC passed?

YES

Water control A



Individual library QC passed?

YES

Pool



QC 2

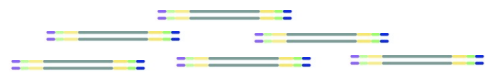
Final QC passed?

YES

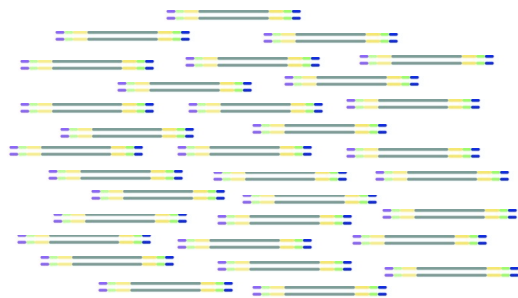


Library prep QC

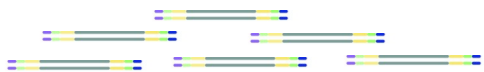
Concentration



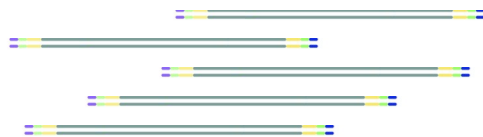
vs



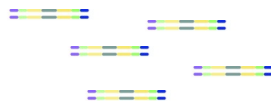
Length



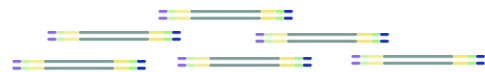
vs



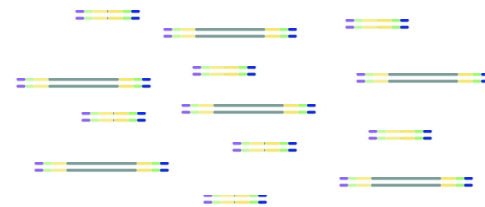
or



Remove adapter dimers



vs



Assessing concentration is important for even pooling and accurate loading

Why is an accurate concentration important?

- **Individual samples:** to pool equal amounts of different samples together
 - Aim for an even amount of sequencing data per sample
- **Final library:** The total amount of final library needs to be within a specific range*
 - For the MinION 10.4 flow cell, this range is 10-50 fmol
 - **We have found that 100-150 fmol works best**

*Nanopore is more forgiving than Illumina in accommodating a wider range of input loading amounts and we find that loading more than 50 fmol works well

Library concentration is best assessed using fluorometry like Qubit

Fluorometry
(e.g. Qubit)

✓ Use for
concentration



- Detects fluorescent dyes that only bind DNA

Library concentration is best assessed using fluorometry like Qubit

Fluorometry
(e.g. Qubit)

✓ Use for
concentration



- Detects fluorescent dyes that only bind DNA

Electrophoresis

(e.g. QSep, Tapestation, Bioanalyzer, gel)



- Compare intensity of DNA bands to standards
- Especially useful to determine concentration in certain fragment length range

✓ Use for size profiling (“quality”) and molar concentrations (for pooling)

Bioanalyzer or Tapestation may be used for pooling, if necessary

Fluorometry (e.g. Qubit)



✓ Use for
concentration

- Detects fluorescent dyes that only bind DNA

Electrophoresis

(e.g. QSep, Tapestation, Bioanalyzer, gel)



- Compare intensity of DNA bands to standards
- Especially useful to determine concentration in certain fragment length range

✓ Use for size profiling (“quality”) and molar concentrations (for pooling)

Nanodrop is not recommended for assessing concentration

Spectrophotometry
(e.g. Nanodrop)



- DNA absorbs light most strongly at 260 nm
- Level of absorbance $\sim$ concentration of DNA

BUT other molecules (RNA, proteins) & contaminants also absorb light at 260 nm, causing up to 10-fold **overestimation** of the DNA concentration!



Do not use for concentration



Use for detecting contaminants

How to use the Qubit

Qubit dsDNA High Sensitivity (HS) Assay kit



- dsDNA HS Reagent (200x concentrate)
- dsDNA HS Buffer
- dsDNA HS Standard #1
- dsDNA HS Standard #2

1. Prepare Working Solution:

- Diluting HS reagent 1:200 in HS buffer

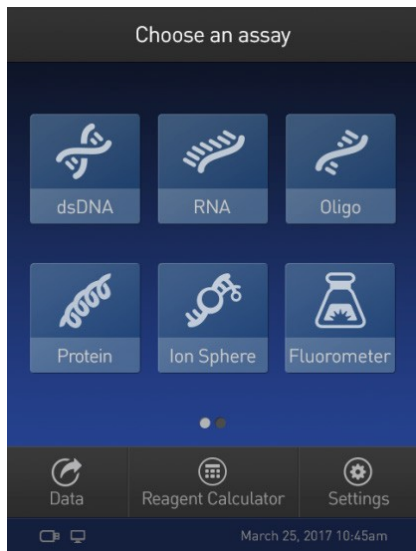
1. Dilute in Working Solution:

- Standards: 10 μ l standard + 190 μ l Working Solution
- Samples: 1 μ l sample + 199 μ l Working Solution

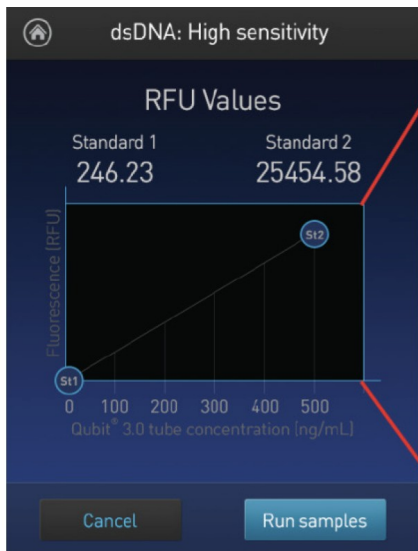
1. Vortex 2-3 seconds before running standards/samples

How to use the Qubit

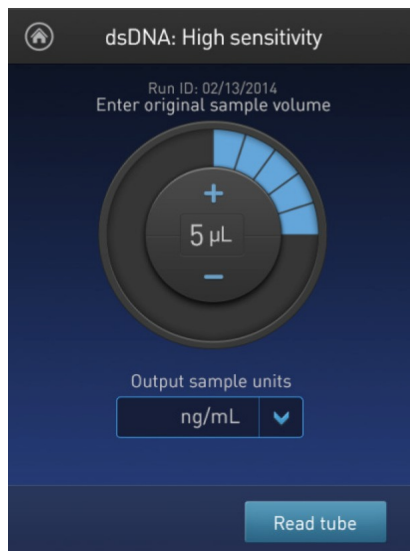
Qubit dsDNA High Sensitivity (HS) Assay kit



Select dsDNA HS workflow



Run the 2 standards



Enter sample volume
added to tube



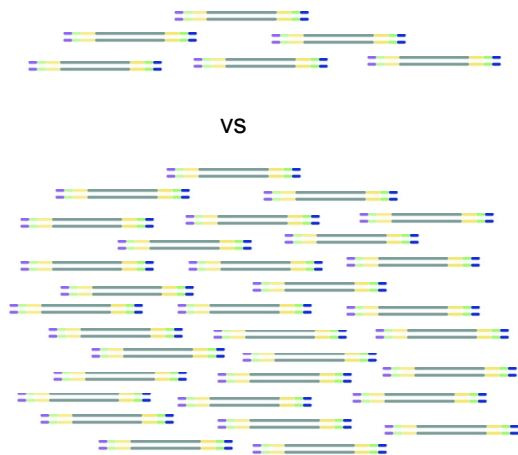
Run sample

Top value = original conc.

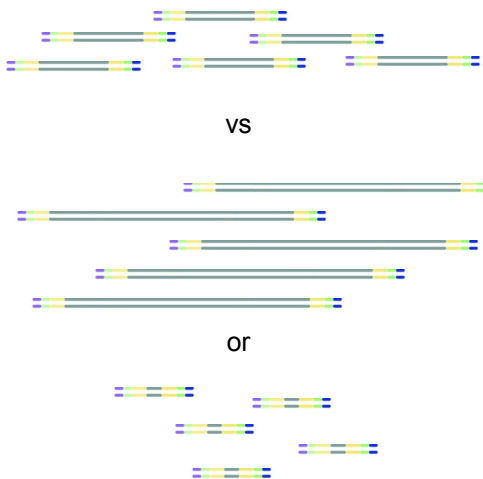
Bottom value = diluted conc.

Library prep QC

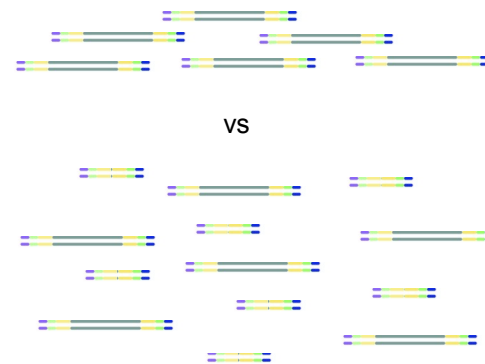
Concentration



Length



Remove adapter dimers



Assessing length can provide insight to the quality of the library:

Is the library good enough to load?

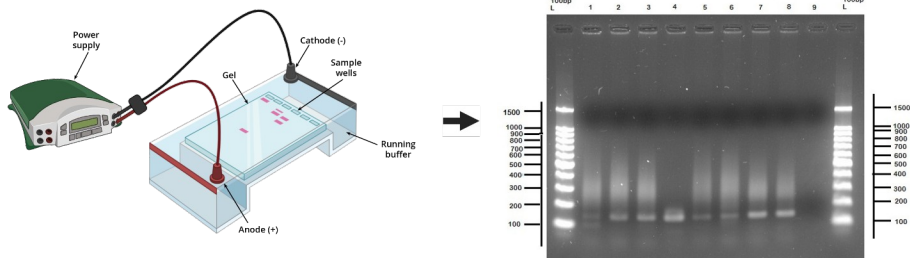
Why is it important to assess library size distribution?

- Depending on sample quality, there can be huge variations in the library size (100s of bases to 1000s of bases)
- Non-specific products can sometimes be formed
 - Primer/oligo dimers when high input cDNA concentrations are used
 - Long concatemers that can block pores and reduce sequencing output (~10kb or more)
- Fragment length distribution impacts molarity calculations for pooling and sequencer loading
 - Convert from ng/μl (Qubit) + Length bp to nM (fmol/μL)

Library size is assessed with electrophoresis methods

How to assess library size distribution?

Gel electrophoresis

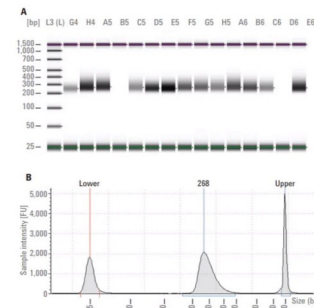


1. PCR amplify an aliquot of the library using “universal” primers
2. Gel electrophoresis of ladder and samples

- Takes ~ 4 hrs
- Does not require specialized equipment
- Tends to slightly overestimate library size

Capillary electrophoresis

E.g. Tapestation, Bioanalyzer



1. Add buffer to aliquot of library
2. Run machine

- Tapestation ~ 10-20 minutes
- Bioanalyzer ~ 45 minutes
- No PCR bias
- Can calculate concentration in nM

Diagnostic PCR method

1. Take 1 uL of the individual libraries or final pool and perform a standard PCR reaction for 20 cycles

Note: the PCR product will only be used for diagnostic QC, not for sequencing

1. For each sample or pool, setup the amplification reaction as follows:

Component	Volume (μL)
LongAmp Master Mix (2x)	25
Nuclease-free Water	23.5
RLB-PCR Primer* (10μM)	0.5
DNA Library Template	1
Total Volume	50

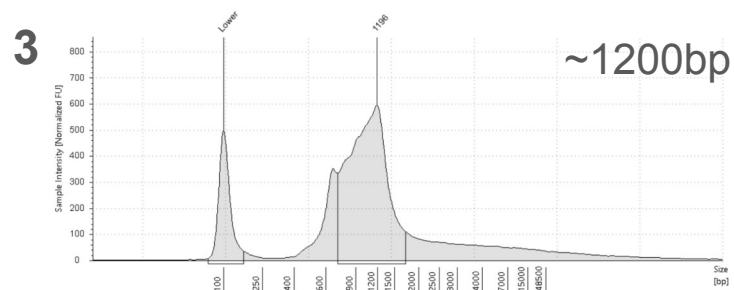
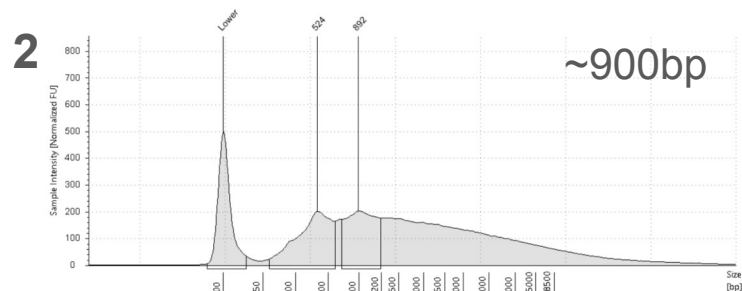
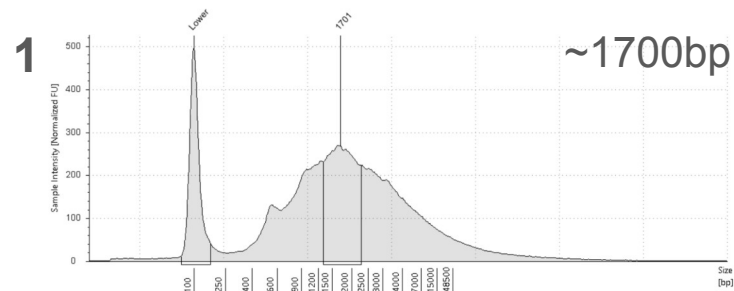
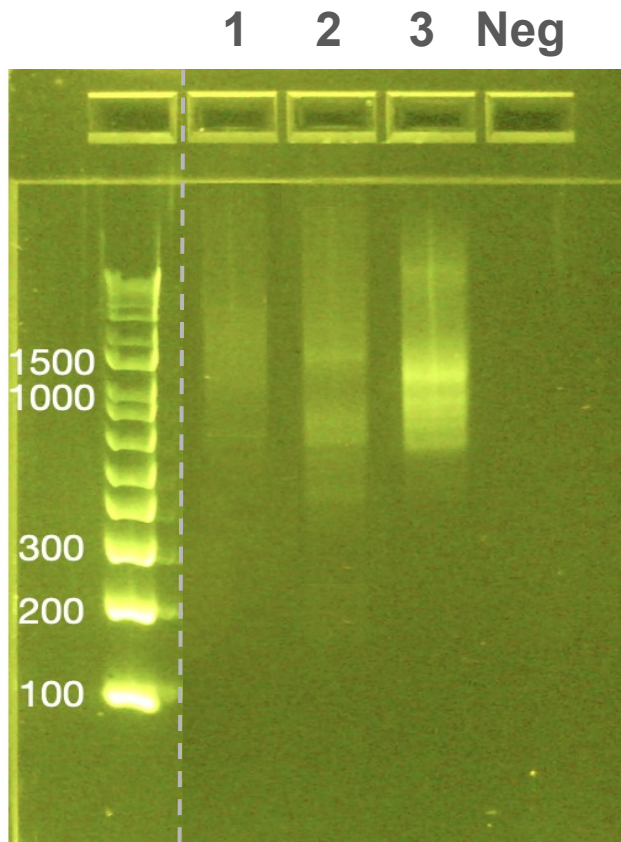
*This primer is only for diagnostic PCR

3. Use the following cycling conditions:

Thermocycler (lid at 105°C)	Cycles
95 °C for 45s	1
95 °C for 15s	20
56 °C for 15s	
65 °C for 5 min	
65°C for 10 mins	1
Hold at 10°C	∞

4. Run PCR amplified product on a gel to visualize and estimate the library size.

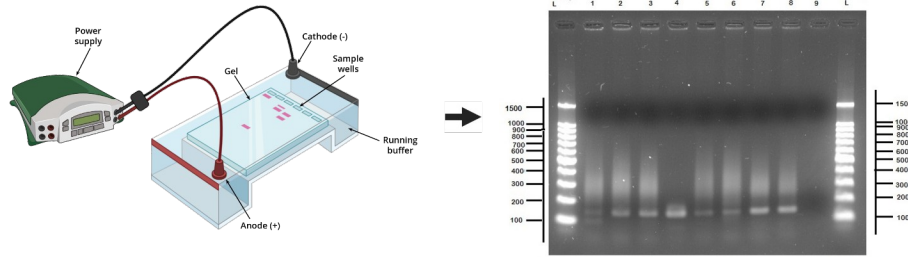
Diagnostic PCR results compared to Tapestation



Library size is assessed with electrophoresis methods

How to assess library size distribution?

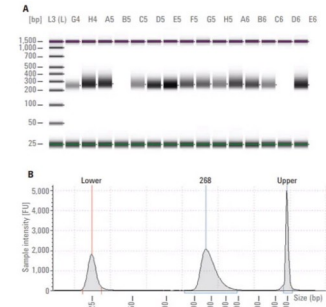
Gel electrophoresis



1. PCR amplify an aliquot of the library using “universal” primers
2. Gel electrophoresis of ladder and samples
 - Takes ~ 4 hrs
 - Does not require specialized equipment

Capillary electrophoresis

E.g. Tapestation, Bioanalyzer



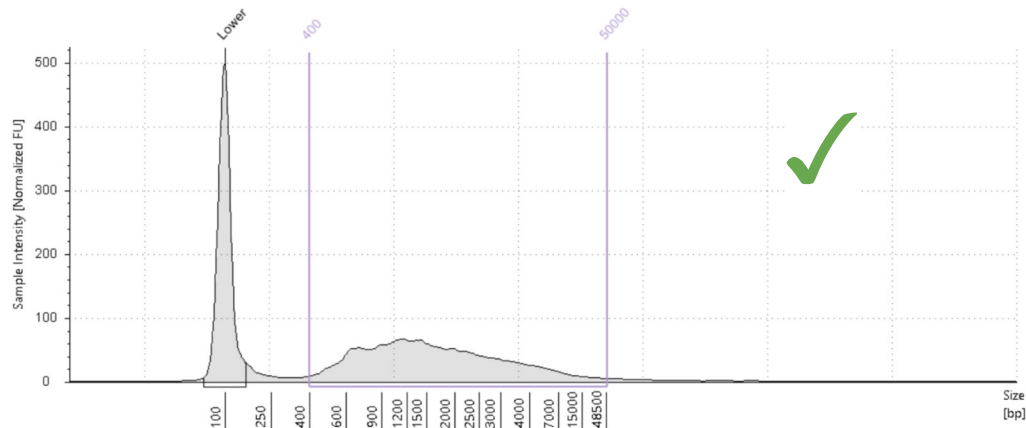
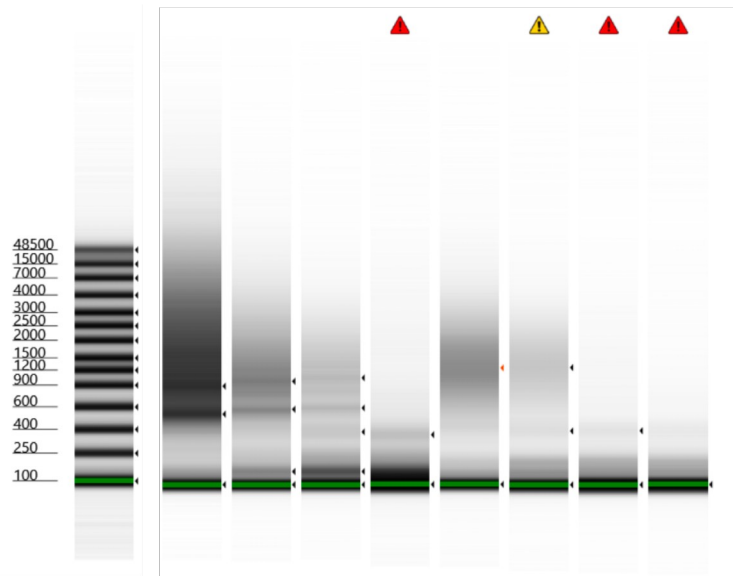
1. Add buffer to aliquot of library
2. Run machine

- Tapestation ~ 10-20 minutes
- Bioanalyzer ~ 45 minutes
- No PCR bias
- Can calculate concentration in nM

Tapestation genomic DNA kit can be used to assess library size

Tapestation

- Genomic DNA kit is most suitable since fragments can be 1000s of bases long



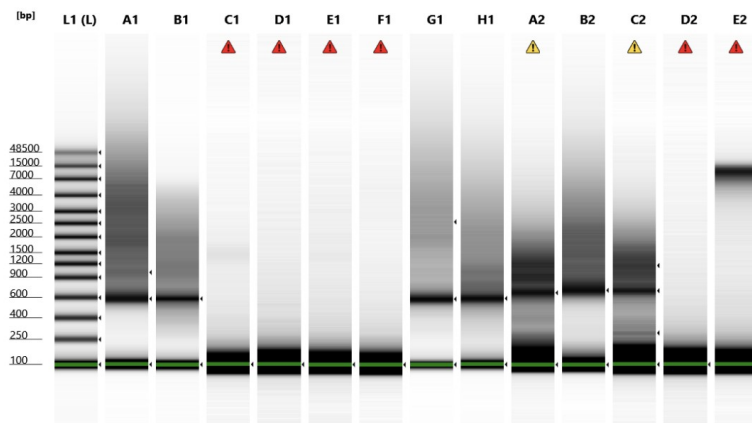
Region Table

From [bp]	To [bp]	Average Size [bp]	Conc. [ng/μl]	Region Molarity [nmol/l]	% of Total	Region Comment	Color
400	50000	2379	12.2	15.0	90.56		

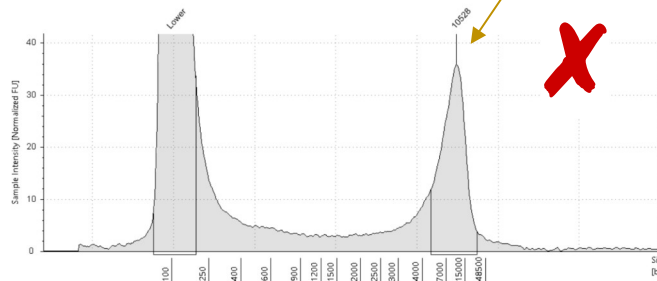
Look out for ~10kb concatemer artifacts - they will clog the flow cell pores

Tapestation

- Genomic DNA kit is most suitable since fragments can be 1000s of bases long



Concatemer peak at ~10kb, with no other peaks*



Sample Table

Well	DIN	Conc. [ng/ul]	Sample Description	Alert	Observations
E2	-	2.39	L13	▲	Sample concentration outside functional range for DIN and the assay

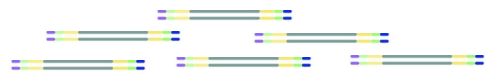
*Consider the input going in, and the library size you would expect. In this case, a 10kb peak was seen for RNA input, which is much longer than the fragment size we'd expect. It turns out that the 10kb peak was a concatemer artifact created during barcoding PCR.

Library size

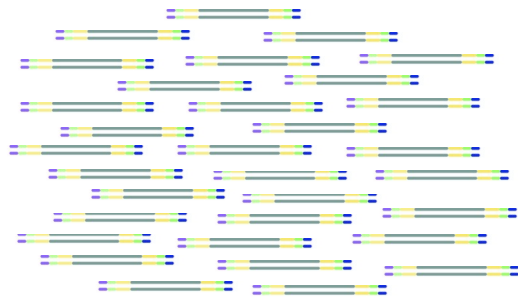
- Checking library size is important for troubleshooting
- It is recommended to routinely check library size when establishing the protocol or expanding to new sample types
- If results are consistent and sample type / source are constant, it is possible for the library size to be estimated
- Size may be evaluated by:
 - Diagnostic PCR + gel electrophoresis
 - QSep / Tapestation / Bioanalyzer / Fragment analyzer
- Size will be used for concentration calculation

Library prep QC

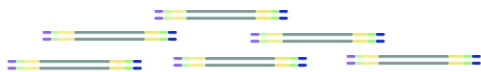
Concentration



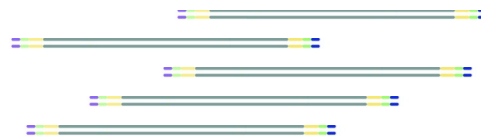
vs



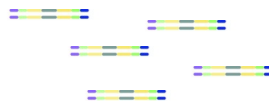
Length



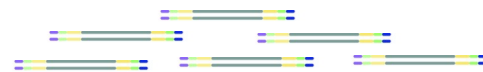
vs



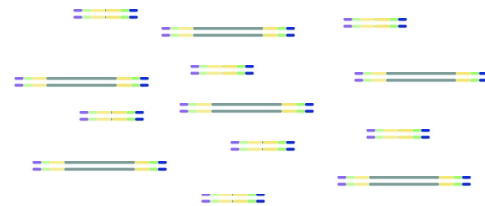
or



Remove adapter dimers

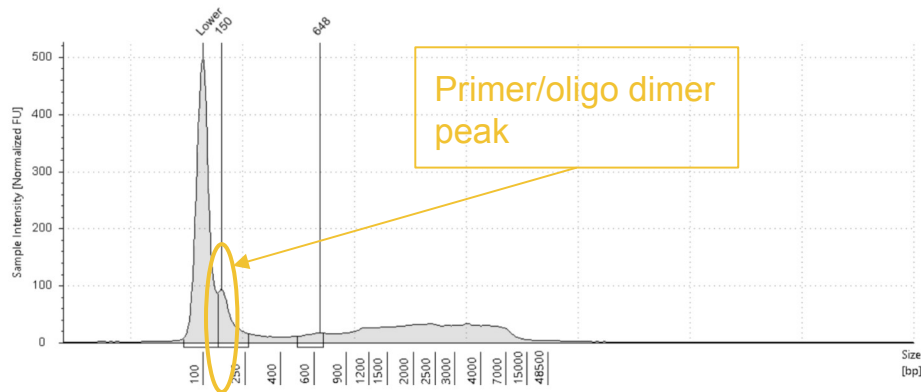


vs



Library prep QC - Clean up primer/oligo dimers

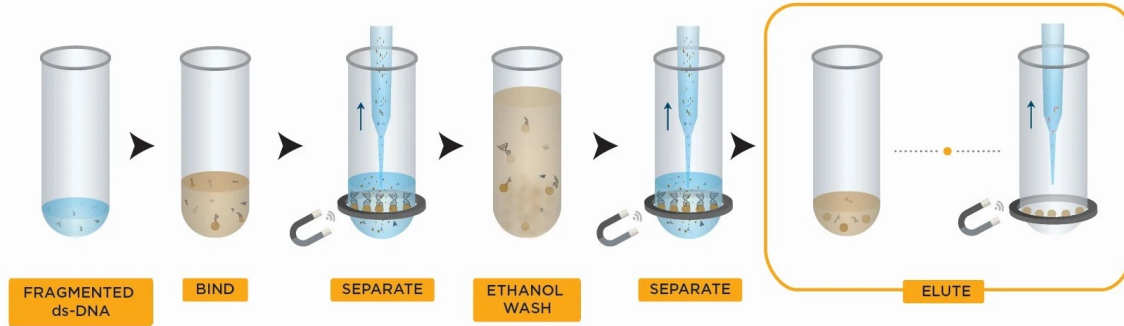
- Primer/oligo dimers:
 - Occur when primers and unused 9N and TSO oligos interact during the PCR amplification
 - Will not be informative if sequenced



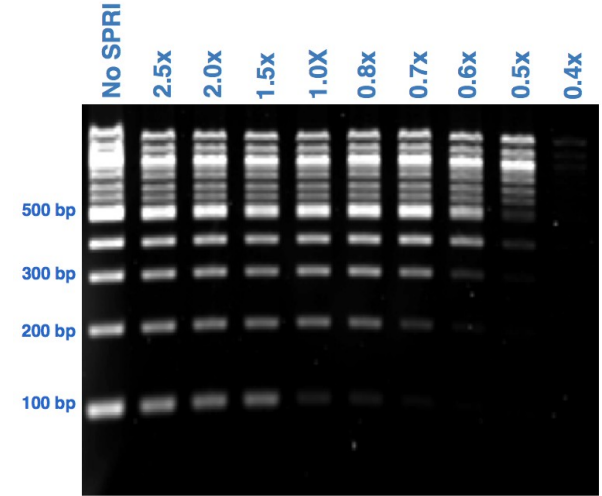
SPRI beads may be used to remove adapters and reagents

SPRIselect or AMPure XP SPRI select beads (paramagnetic)

- The longer a fragment, the better it will bind to the beads
- The fewer beads you add, the more short fragments you'll lose



1. Bind DNA to magnetic beads; 2. Separate beads from contaminants;
3. Wash magnetic beads with 85% ethanol to remove contaminants;
4. Elute DNA from magnetic beads; 5. Transfer sample to new tube



$$\text{SPRI ratio} = \frac{\text{Volume beads}}{\text{Volume reaction}}$$

Example: for a 1.8x SPRI clean-up:

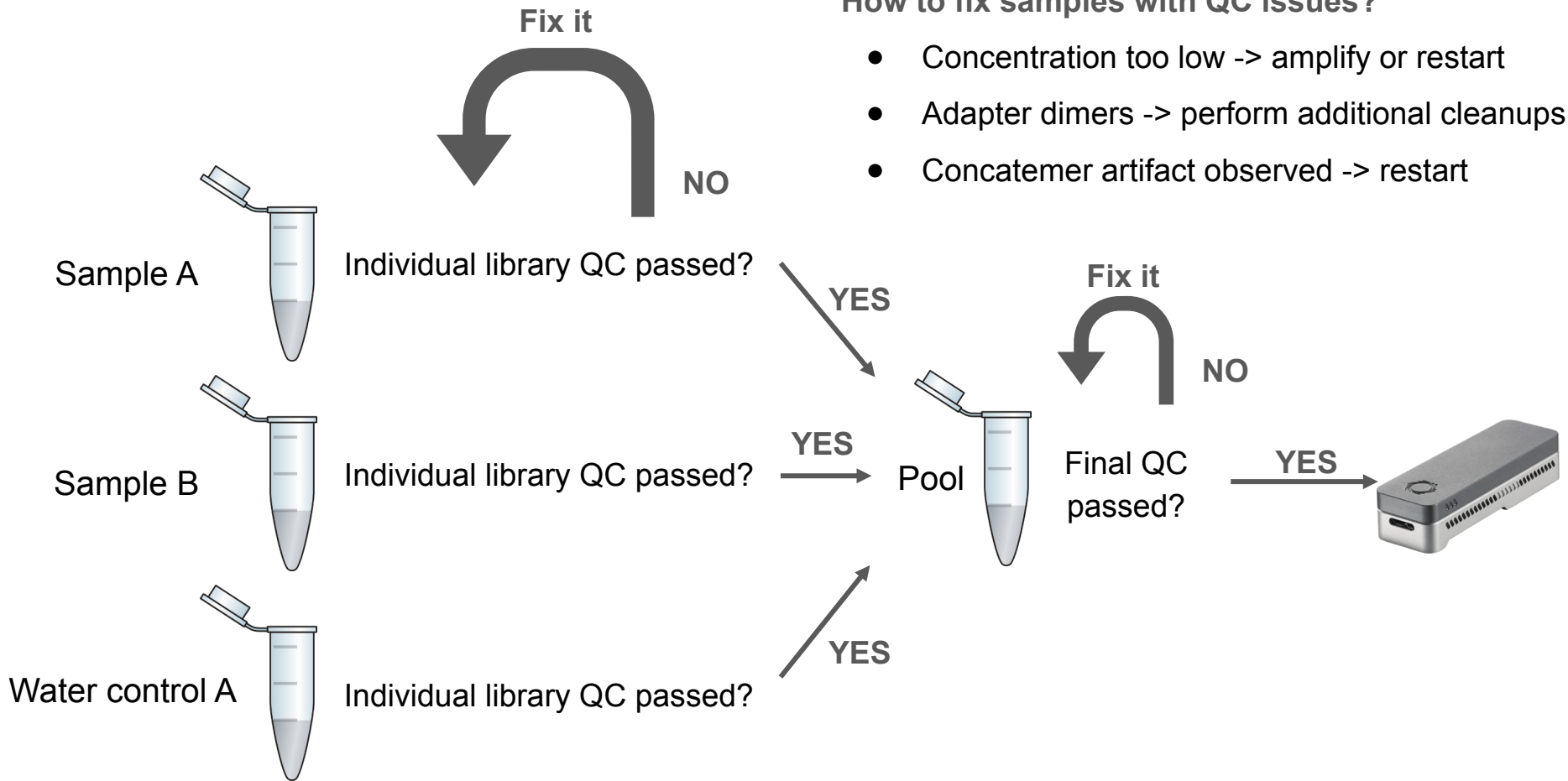
- Add 18 μl of SPRI beads to 10 μl of sample

Library prep QC - Clean up adapter dimers

AMPure XP SPRI select beads (paramagnetic)

- Size selection using SPRI beads to remove adapter dimers
- Samples can first be pooled and then a 0.6x SPRI clean-up can be performed

Library prep QC



How much concentration do I really need?

- Final pool
 - Need 100-150 fmol to load the flow cell
 - = 11ul of 10-15nM
- Individual samples:
 - Ideally individual samples would also have 10-15nM
 - (7-10ng/ul for 1000bp library)
 - (13-20ng/ul for 2000bp library)
 - Total mass (ng) per sample added to pool will depend on:
 - library size
 - # of samples in the pool